

Learning from examples: Machine Learning

BS-AI & MMG

Faculty of Emerging Sciences & Technologies

Agenda

Learning from Examples

Introduction to Machine Learning

Importance of Machine Learning

Types of Machine Learning

Applications of Machine Learning

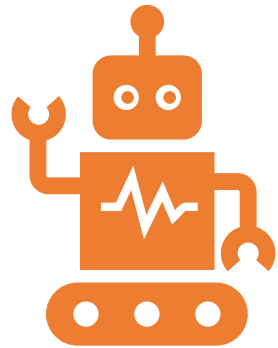
Machine Learning: State-of-the-art

Data Mining versus Machine Learning

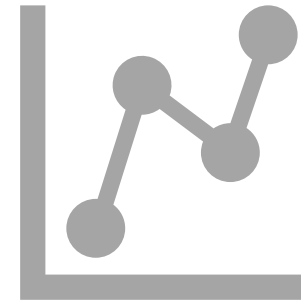
Role of Data Analyst and Machine Learning Engineer

Recap of the Lecture

Learning from Examples



A machine **improves its performance** through experience.



The process involves **observing data**, building a **model**, and **using it to predict or act**.



Why Machines Should Learn?

Why Machines Should Learn?

- **Two main reasons:**
 - **Unpredictable environments:** Machines face new, unseen situations (e.g., self-driving cars, stock prediction).
 - **Implicit human skills:** Some human abilities (like recognizing faces) cannot be directly programmed.

Why Machines Should Learn?

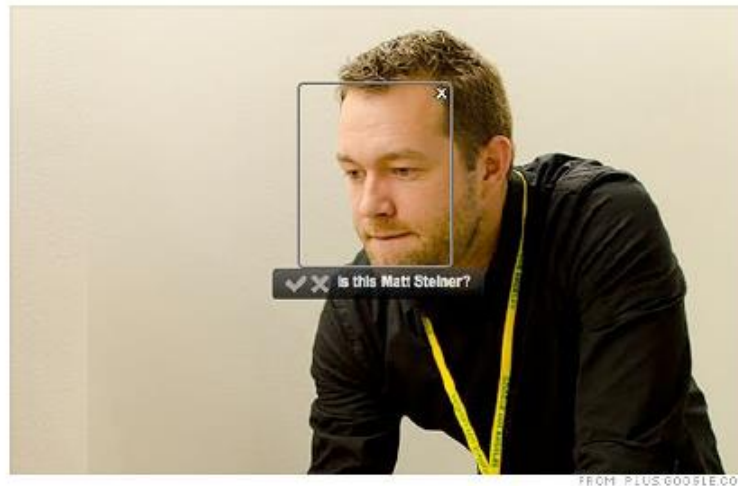
- Certain tasks are extremely difficult to program by hand:



Spam filtering

Why Machines Should Learn?

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Spam filtering
Face recognition

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Spam filtering
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Machine translation

Why Machines Should Learn?

- Certain tasks are extremely difficult to program by hand:

«hi! how are you doing?»

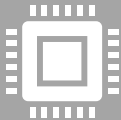


Spam filtering
Face recognition
Machine translation
Speech recognition

Background of Machine Learning



Arthur Samuel — an American pioneer in artificial intelligence and computer gaming was the first to define **machine learning** in 1959 as:

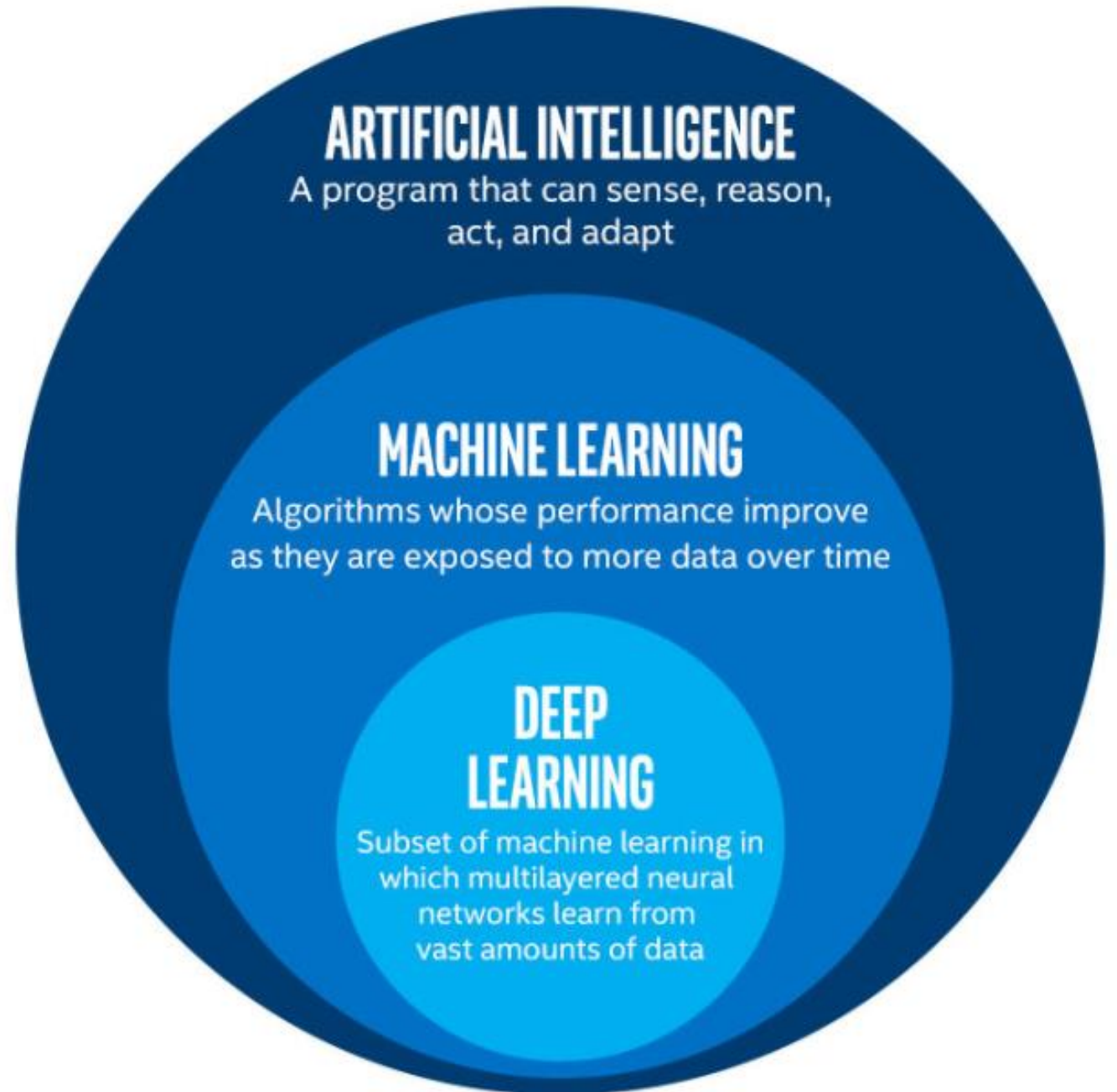


“The field of study that gives computers the ability to learn without being explicitly programmed.”



He developed one of the earliest self-learning programs — a **checkers-playing program** — which improved its performance over time by playing against itself.

Machine Learning



Machine Learning

- Closely related to data mining and statistics

Data Mining:

Concerned with the analytics of data.

Statistics:

Concerned with prediction-making/probability.

Machine Learning

- **Machine learning** is an application of artificial intelligence that automates analytical model building by using algorithms that iteratively learn from data without being explicitly programmed.
- A system to ask questions and get answers.



Machine Learning

- The study and construction of algorithms that can learn from and make predictions on data

What is Machine Learning

- Machine Learning is the field of study that gives computers the ability to learn without being explicitly programmed.
- A computer program is said to learn from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E .
- Machine learning is programming computers to optimize a performance criterion using example data or past experience.

What is Machine Learning

- Suppose your twitter program watches which tweet you do or do not mark as cyberbullying, and based on that learns how to better filter bullying tweet. What is the task T in this setting?
 - a. Classifying tweets as bully or no bully.
 - b. Watching you label tweets as bully or no bully.
 - c. The number (or fraction) of tweets correctly classified as bully or no bully.
 - d. None of the above—this is not a machine learning problem.

Generalities

- Virtually all learning problems can be formulated as (complex) mappings between inputs and outputs
- We are trying to learn what is the best output $\mathbf{o}$ to produce for each possible input $\mathbf{i}$
- Mathematically speaking, we search for a «good» function $\mathbf{F}: \mathbf{I} \rightarrow \mathbf{O}$, where $\mathbf{I}$ is the set of possible inputs, and $\mathbf{O}$ the set of possible outputs

Generalities - Examples

	Input i	Output o
Spam filtering	An email	{spam, non-spam}
Face recognition	An image	Identified faces
Machine translation	A sentence in language A	A sentence in language B
Speech recognition	A speech signal	A (text) sentence
Data mining	A financial transaction	{fraud, non-fraud}
Robot motion	Sensory data	Motor control

Learning Methods

- But how do we learn this mapping?
- The learning method depends on the kind of data that we have at our disposal
 - We can have examples of data where we have both the inputs and outputs: $(\mathbf{i}, \mathbf{o})$ 
 - For some data, we only have the inputs $\mathbf{i}$ 

Learning Methods

- But how do we learn this mapping?
- The learning method depends on the kind of data that we have at our disposal
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supervised learning
 - For some data, we only have the inputs $\mathbf{i}$ 

unsupervised learning

Types of Machine Learning

Supervised

- Regression
- Classification

Unsupervised

- Clustering

Reinforcement Learning

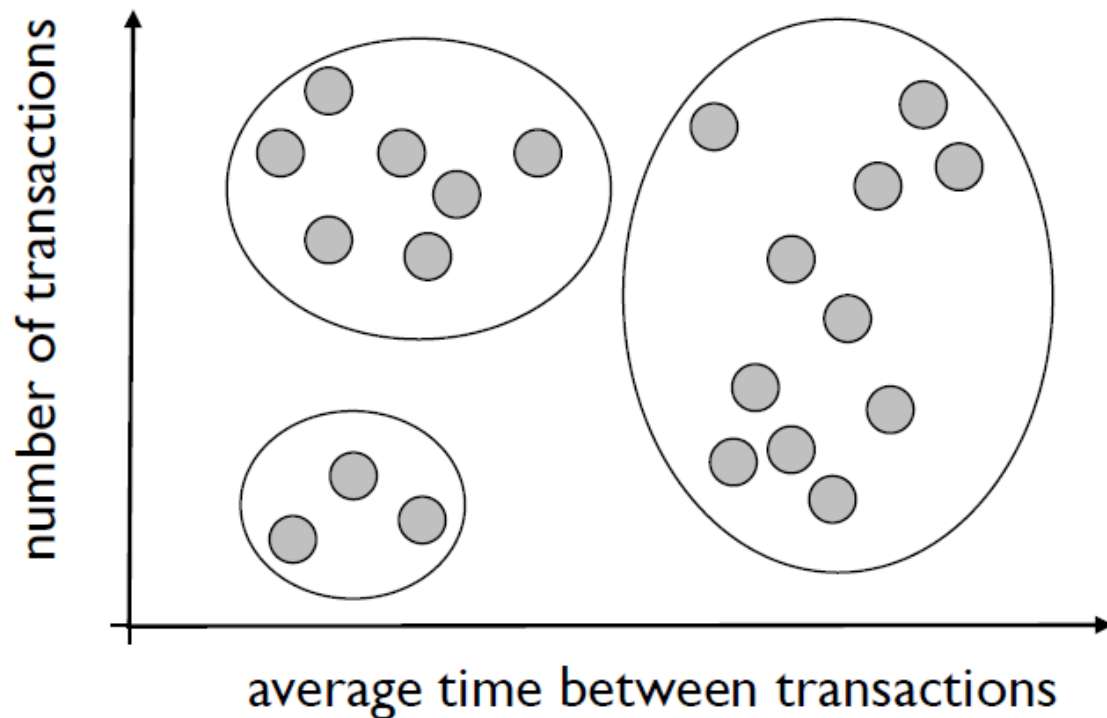
Supervised Learning

- In supervised learning, we have *training data* encoded as pairs (i,o) , where the «correct» output is often manually annotated
 - E.g. spam filtering, machine translation, face recognition, etc.
- The function $\mathbf{F}: \mathbf{I} \rightarrow \mathbf{O}$ is often dependent on a (sometimes large) set of parameters
- ... and the learning goal is to «adjust» these parameters in order to fit the data

Supervised Learning

- A good learning model is a model that **generalises** well to new data
 - In other words, it is able to *abstract* over its experience to detect underlying patterns
 - The design and test of such models is a crucial part of machine learning
- Else, the model is said to be *overfitted*
 - In other words, it is very well «fitted» to the examples it has processed, but perform very poorly with unseen data

Unsupervised Learning



- Sometimes, we don't have access to any output value $\mathbf{o}$, we simply have a collection of input examples $\mathbf{i}$
- In this case, what we try to do is to learn the *underlying patterns* of our data
 - is there any *correlations* between features?
 - can we *cluster* our data set in a few groups which behave similarly, and detect *outliers*?

Unsupervised Learning

- In unsupervised learning, the model is trained on **unlabeled data**.
- The model tries to find underlying patterns or structures in the data without being explicitly told what to predict.
- **Key Features:**
- **Training Data:** Contains only input data without output labels.
- **Learning Process:** The model looks for patterns or groupings in the data (such as clustering similar data points).
- **Applications:** Clustering, anomaly detection.

Examples of Unsupervised Learning Algorithms:

- **K-Means Clustering** (for clustering tasks)
- **Hierarchical Clustering**
- **Autoencoders** (for anomaly detection)

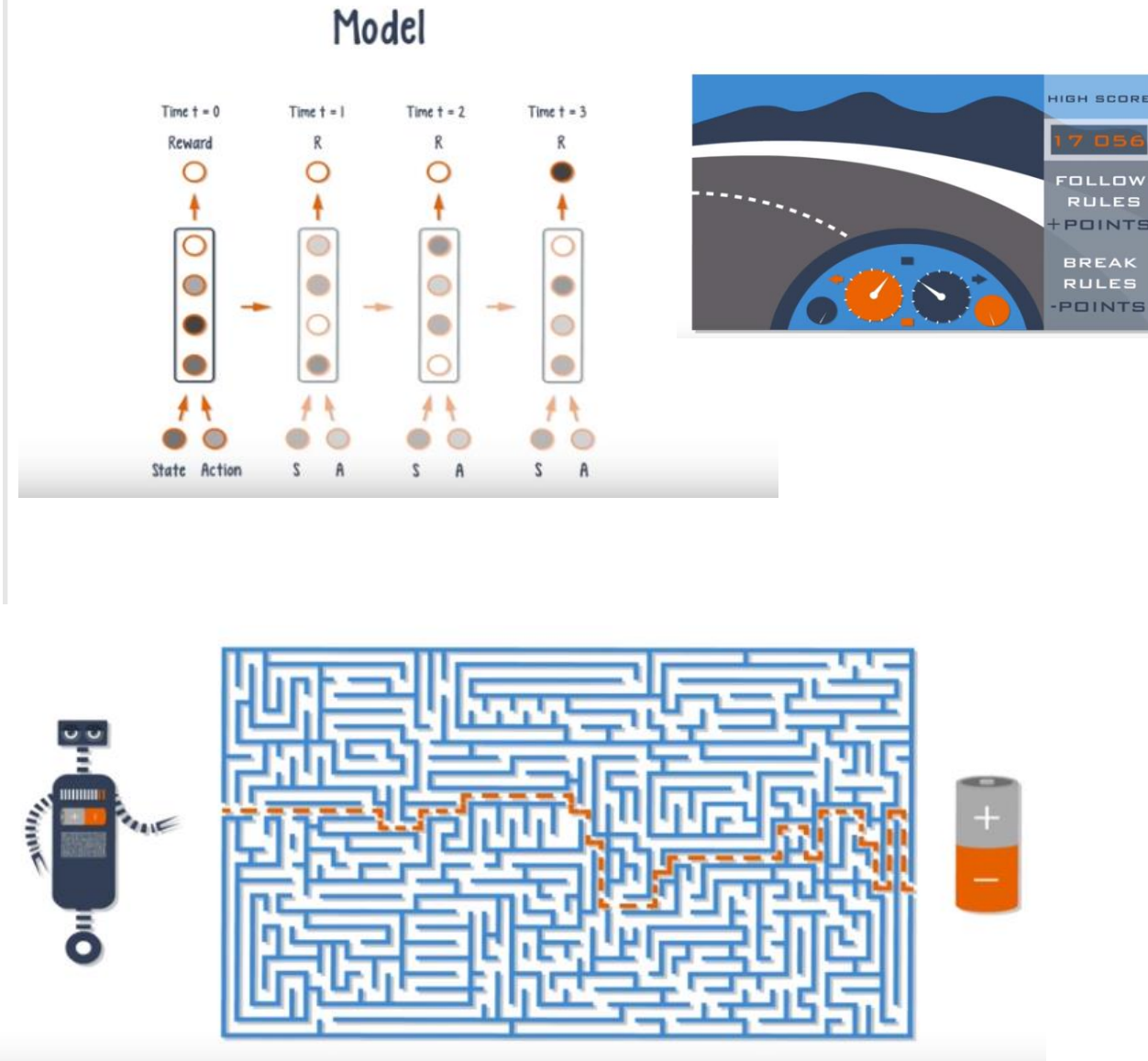
Example Use Cases:

Customer segmentation (grouping customers based on purchasing behavior for targeted marketing)

Anomaly detection (identifying unusual patterns, like fraud detection)

Reinforcement Learning

- Involvement of AI, animal psychology, and control theory
- Reinforcement learning (RL) can learn from experience and interaction with the environments.
- Agent interacts with environment and its goal is to maximize the numerical reward



Growth of Machine Learning

Machine
learning is
preferred
approach
to

- Speech recognition
- Natural language processing
- Computer vision
- Robot control
- Computational biology

Data Mining versus Machine Learning

Data Mining is used to look for emerging patterns that can help shape our decision-making processes.

For example:

- data to explore best-selling items,
- what was returned the most,
- customer feedback to help sell more clothes and enhance product recommendations.

This use of data analytics can lead to an improved customer experience overall.

Data Mining versus Machine Learning



Machine learning, on the other hand, can actually learn from the existing data and train that machine to teach itself.

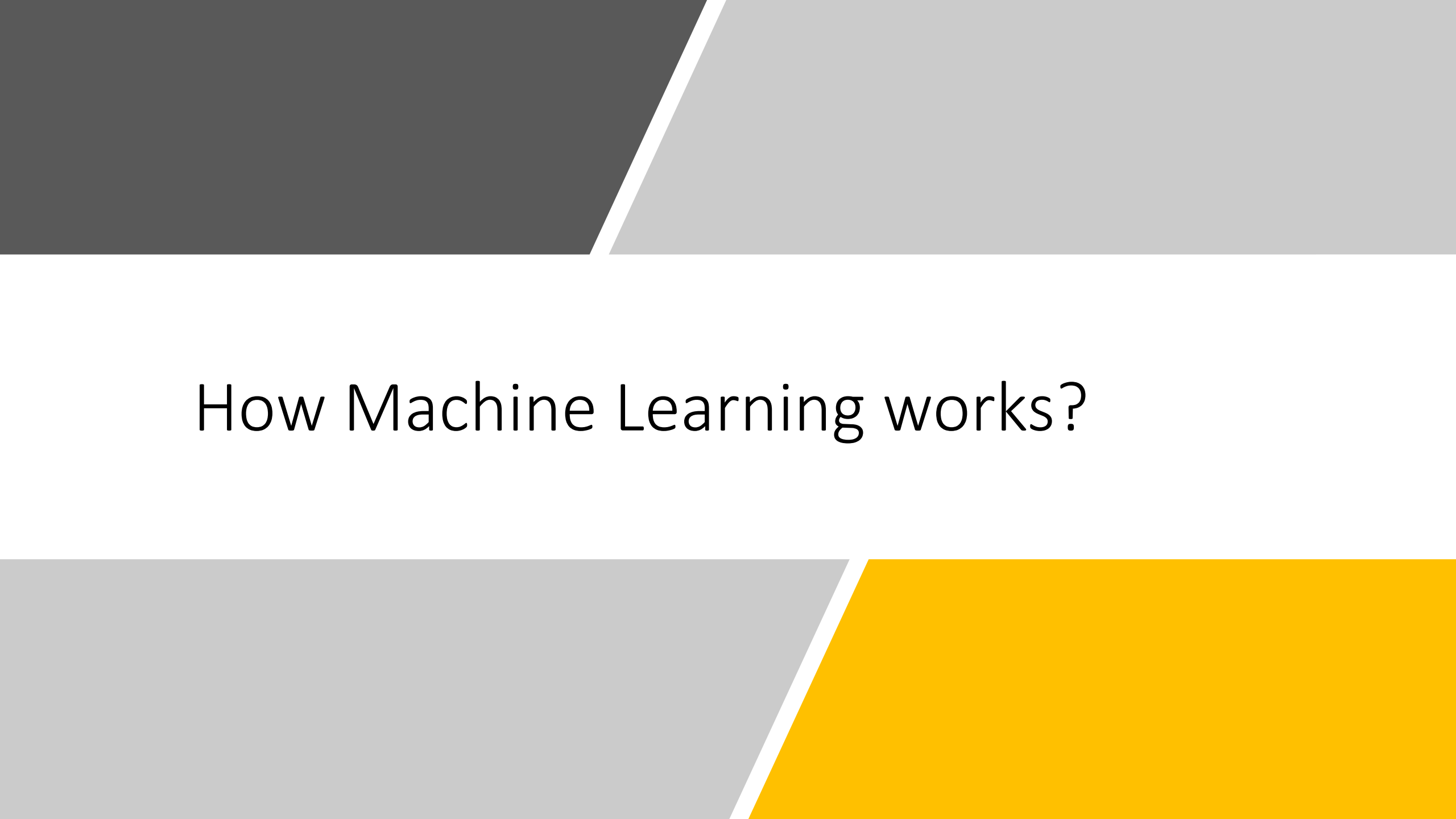


For example: Ham or Spam email



Machine learning can look at patterns and learn from them to adapt behavior for future incidents,

while data mining is typically used as an information source for machine learning to *pull* from.



How Machine Learning works?

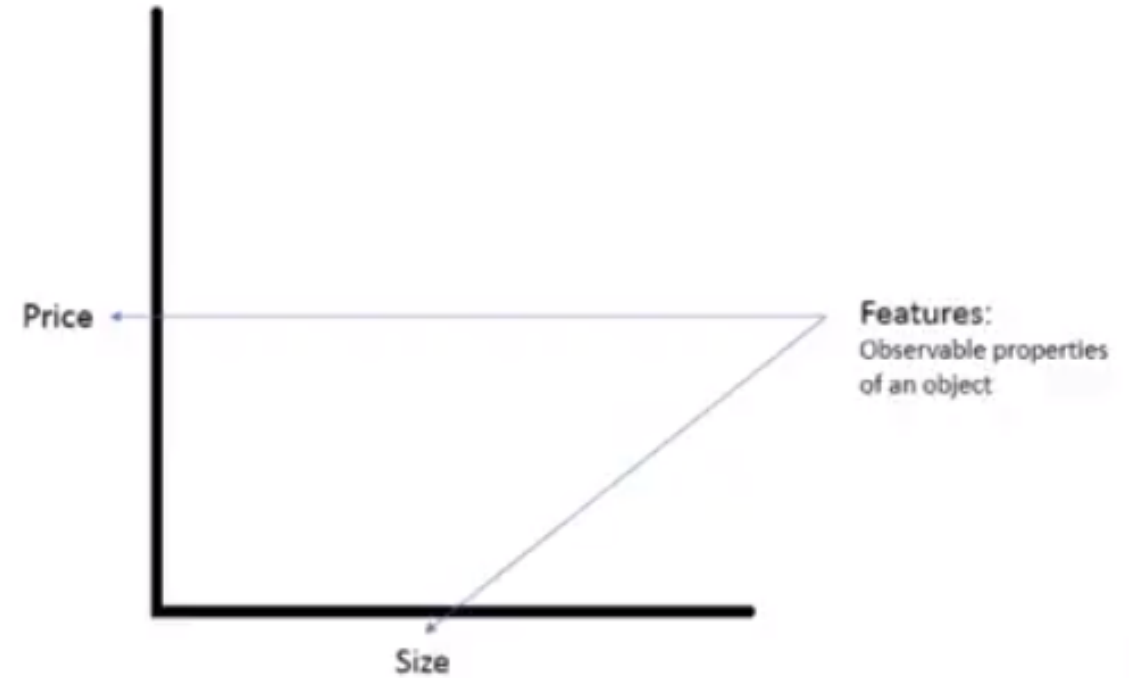
Example

The size and price of house.



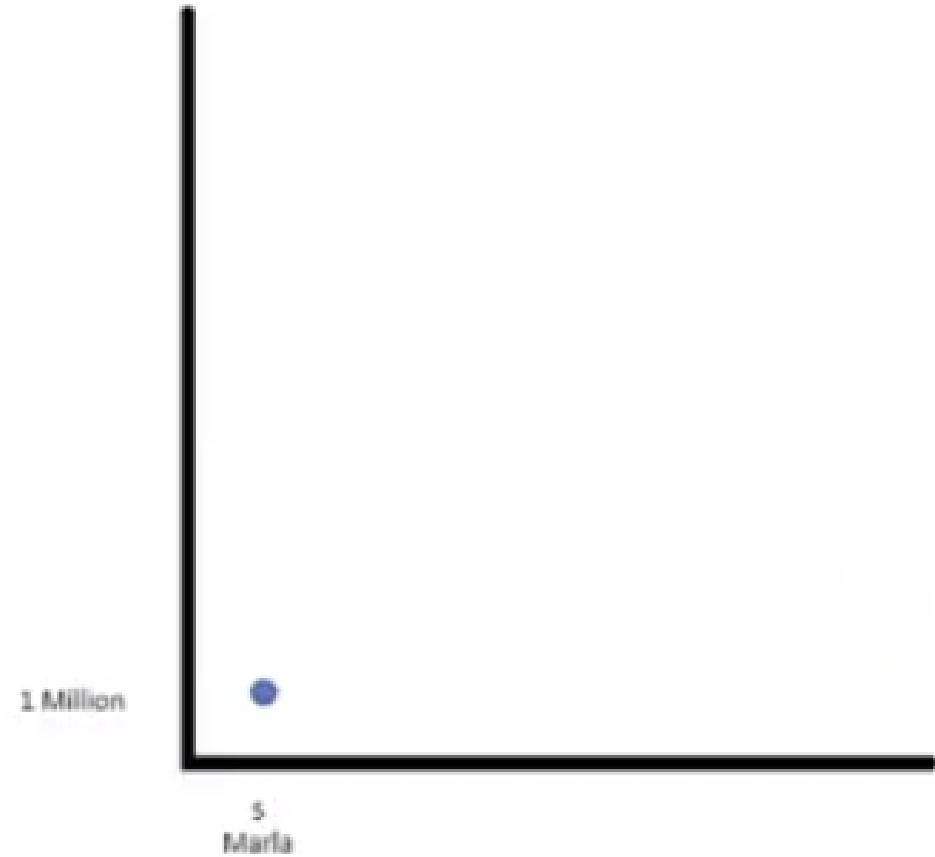
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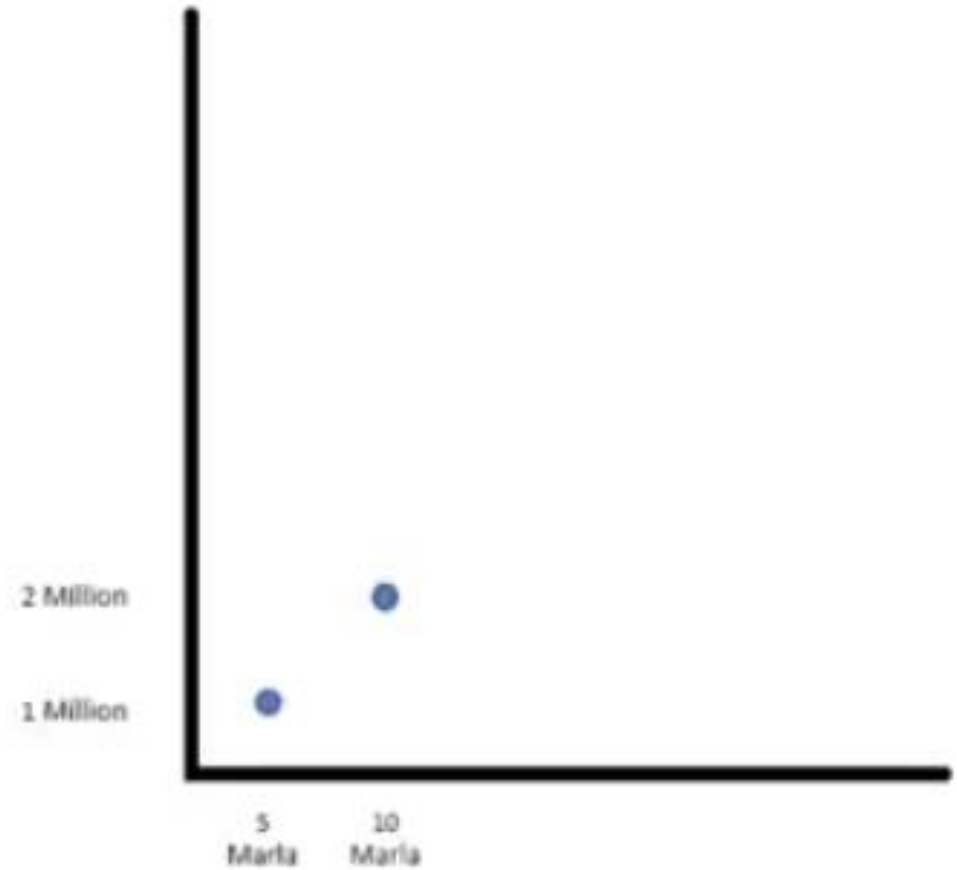
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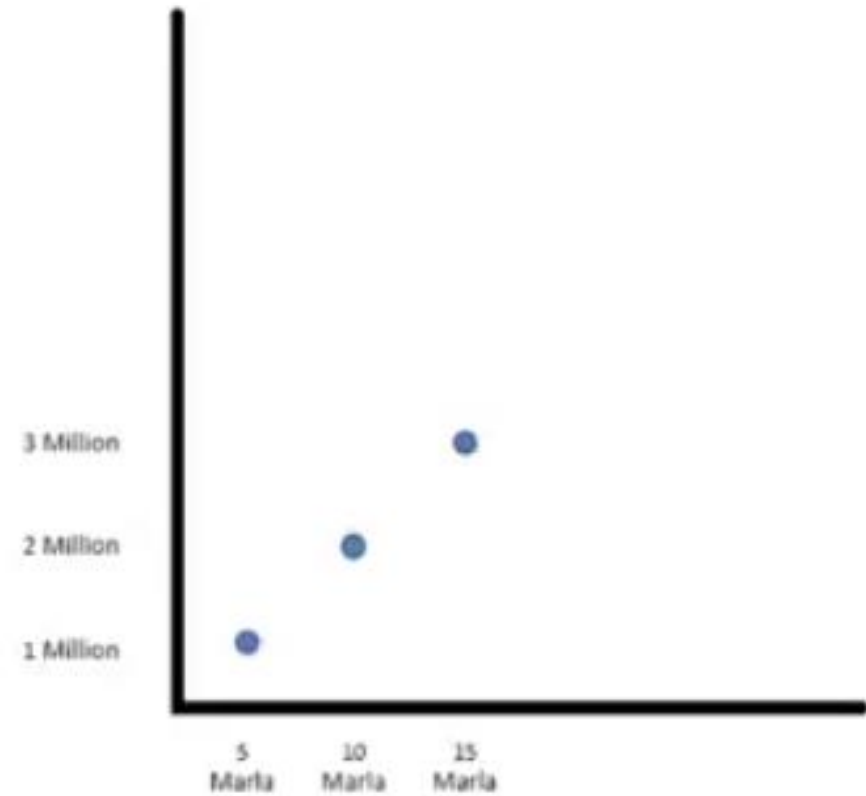
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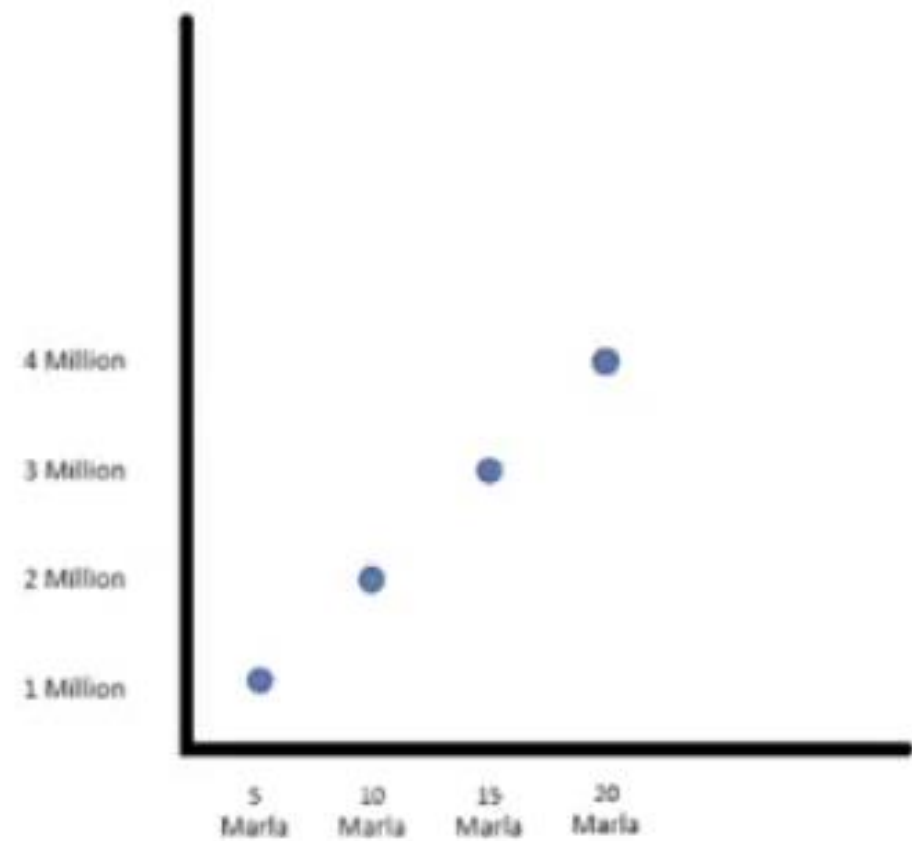
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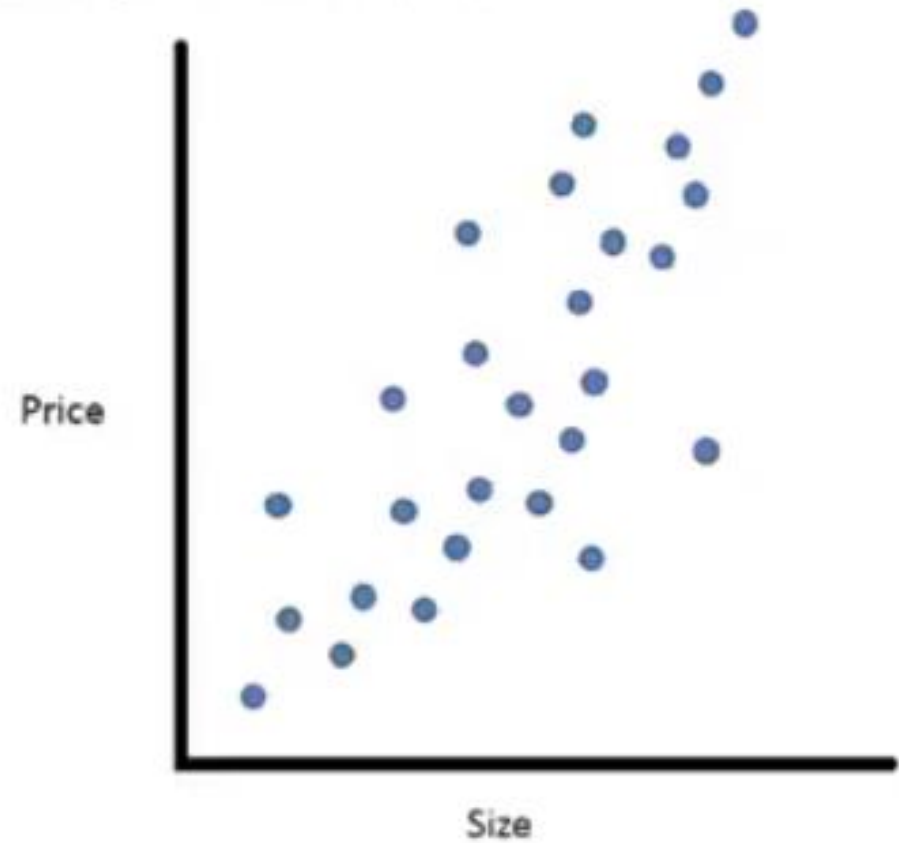
Example

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Example

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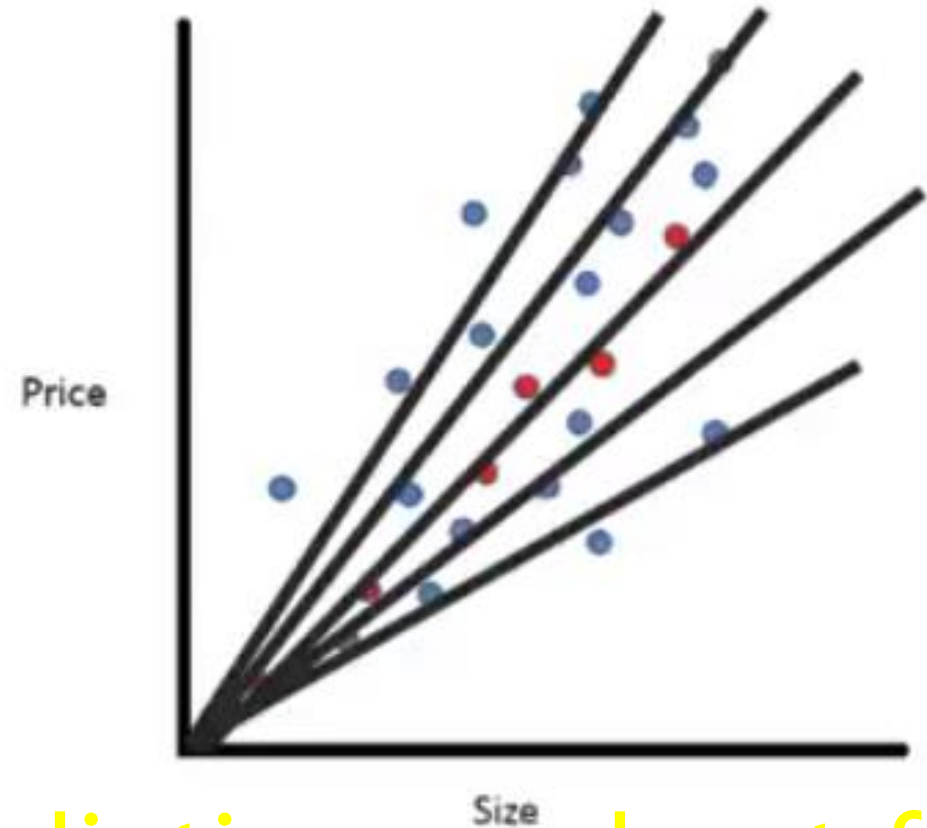


$$y = m * x + b$$

x = input
y = output
m = slope
b = y intercept

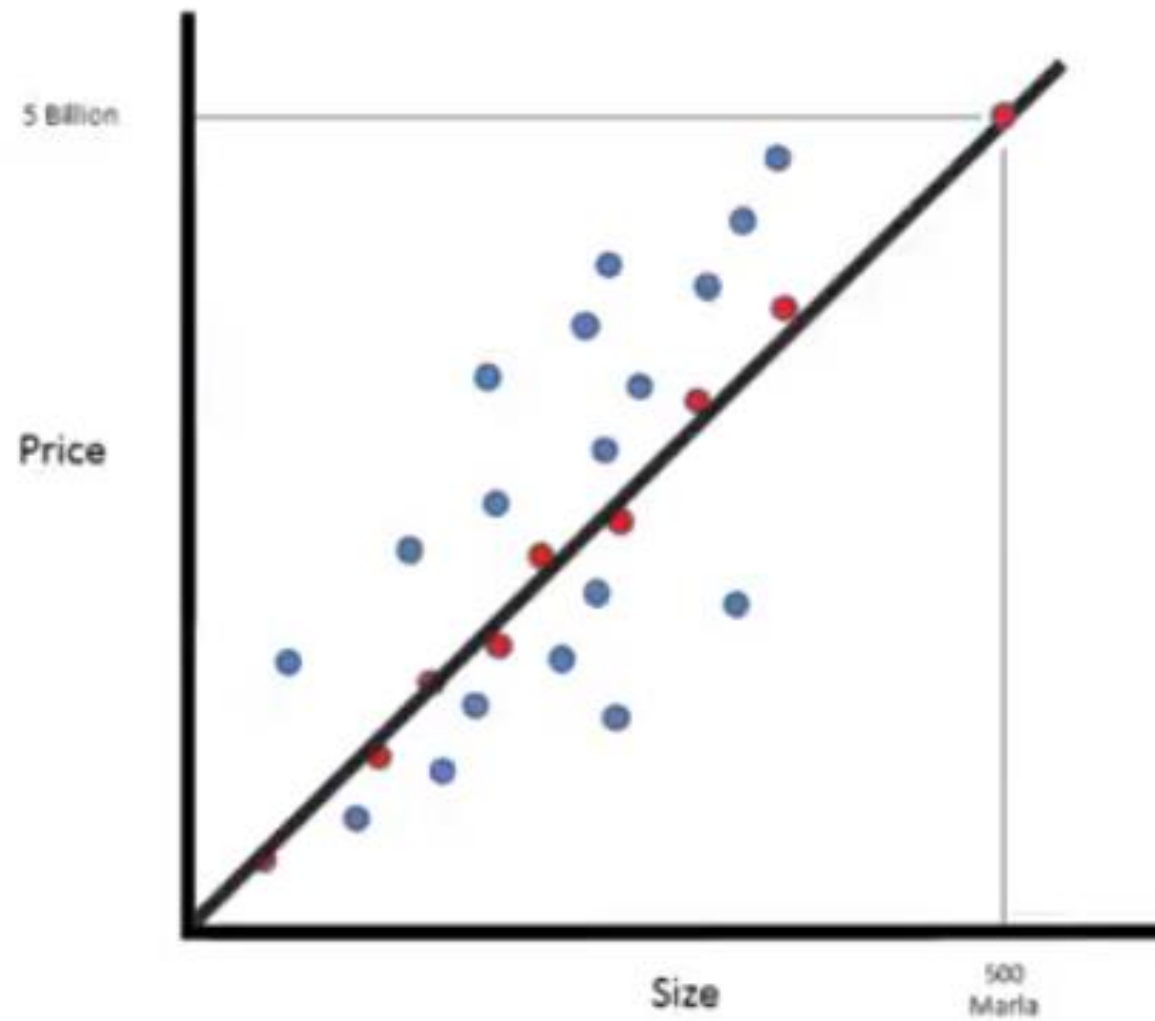
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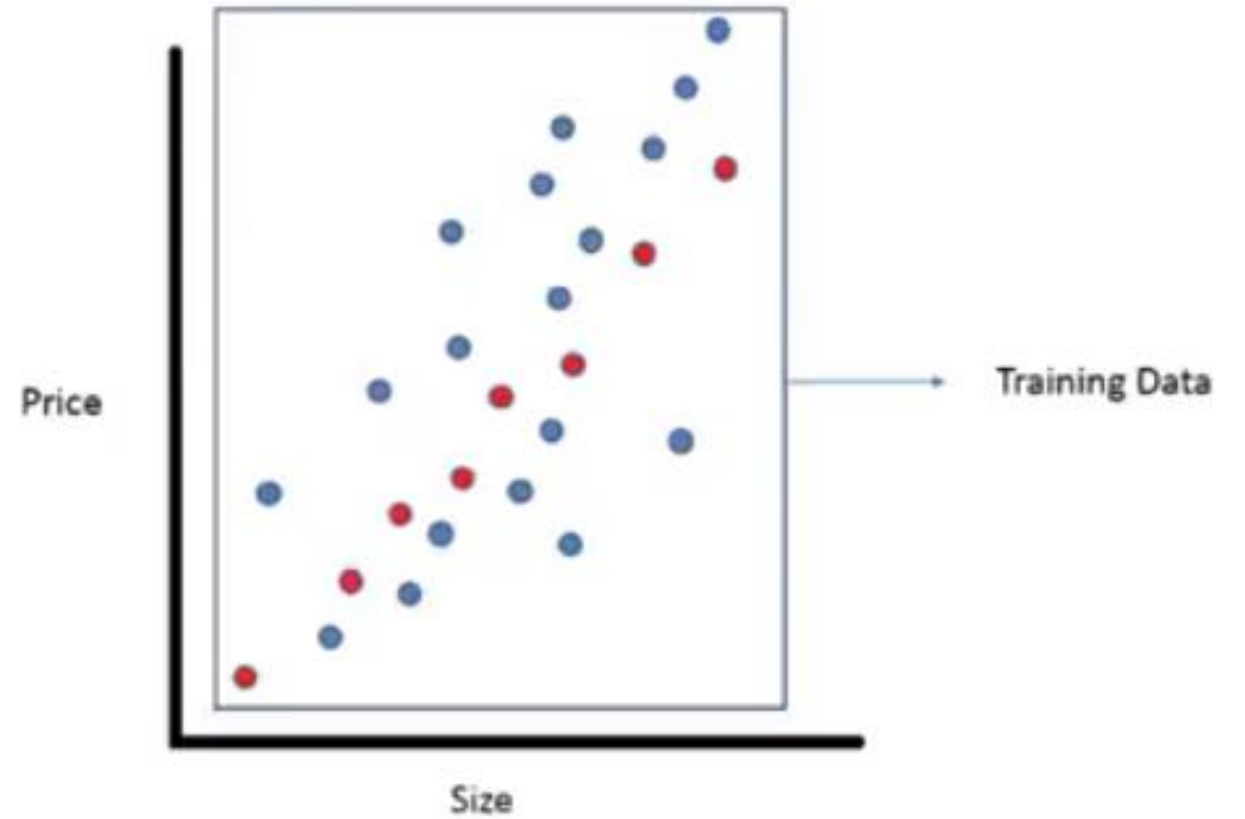


Prediction on best fit
line

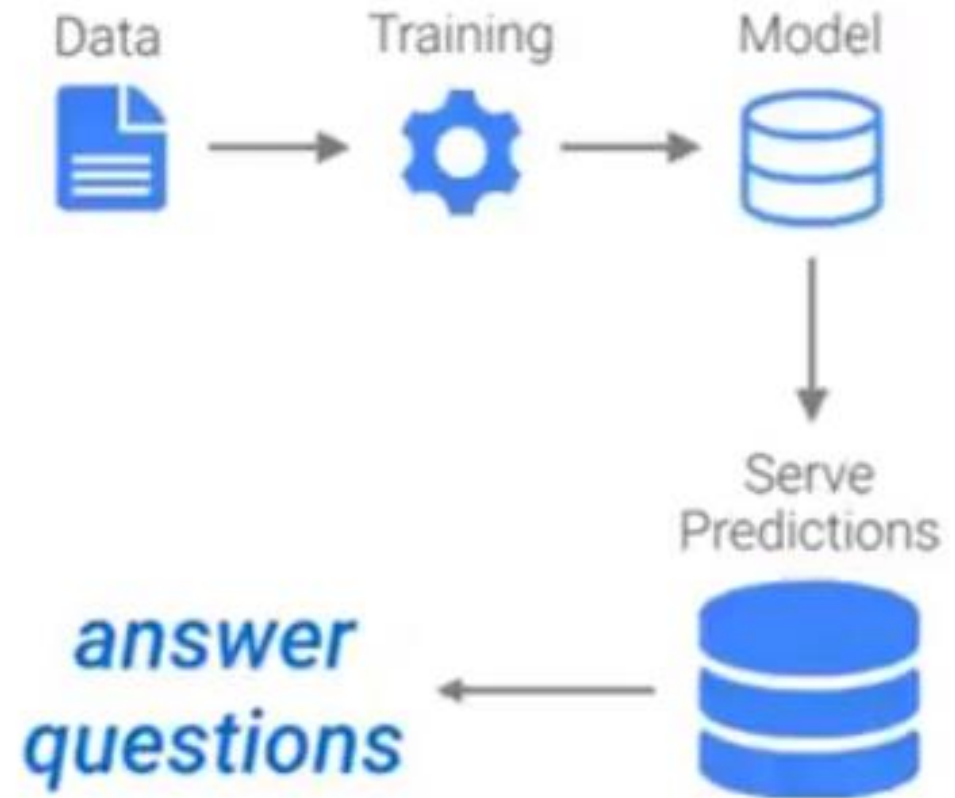
Example



Training data



Process of training and prediction or ML

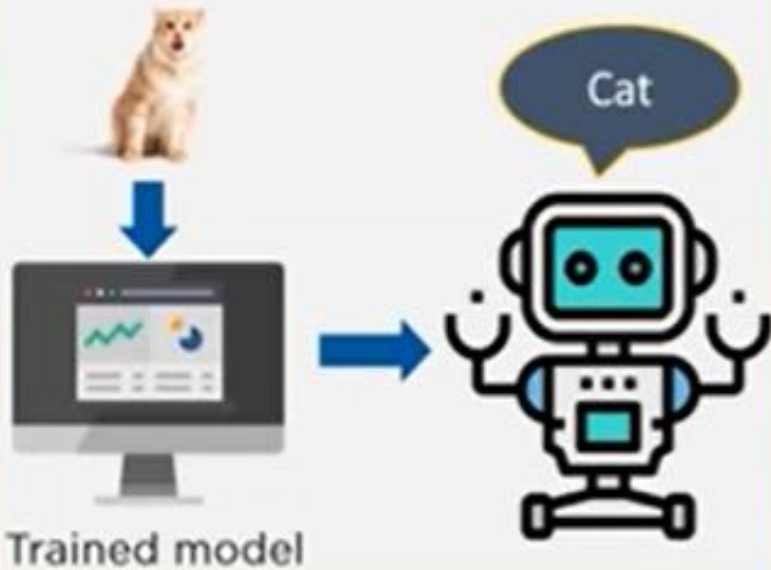


Difference b/w Supervised,
Unsupervised &
Reinforcement Learning?

Definition

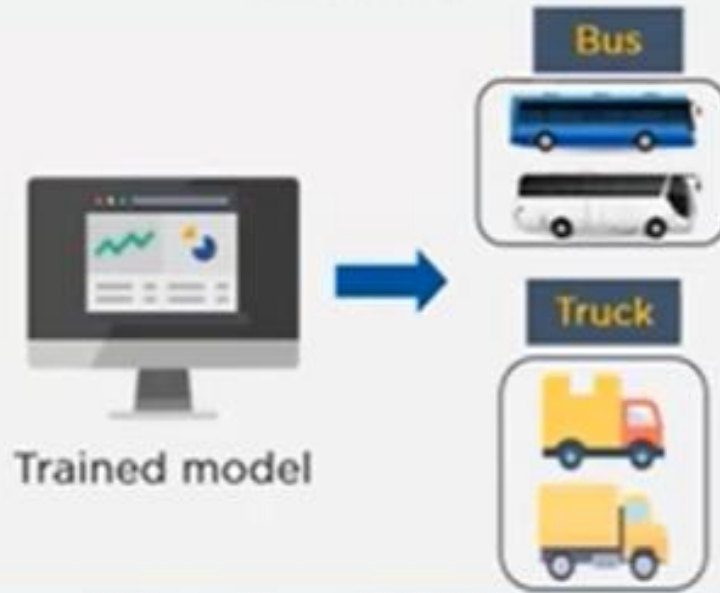
Supervised Learning

Supervised Learning is used to train machines using labeled data



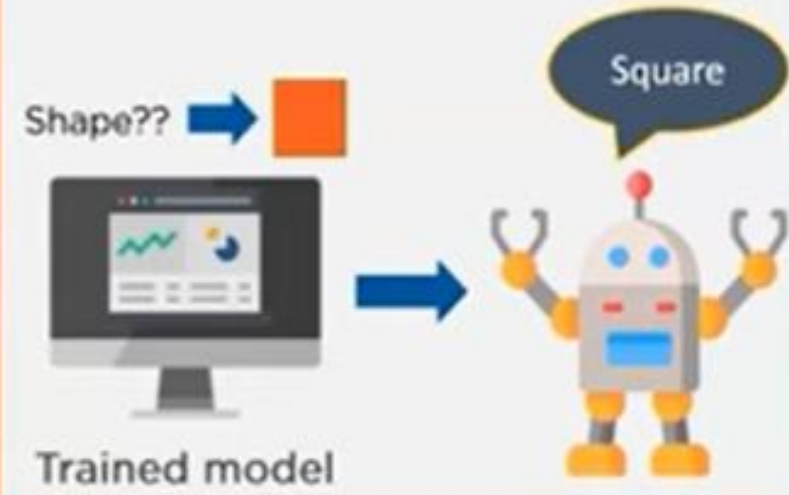
Unsupervised Learning

Unsupervised Learning uses unlabeled data to train machines



Reinforcement Learning

Reinforcement Learning uses an agent and an environment to produce actions and rewards



Algorithms

Supervised Learning

Linear Regression

Logistic Regression

Support Vector Machines

K Nearest Neighbors

Decision Tree

Unsupervised Learning

K Means Clustering

Hierarchical Clustering

DBSCAN

Principal Component
Analysis

Reinforcement Learning

Q-Learning

SARSA

Monte Carlo

Deep Q Network

Approach

Supervised Learning

Takes labeled inputs and maps it to the known outputs



Unsupervised Learning

Understands patterns and trends in the data and discovers the output



Reinforcement Learning

Follows trial and error method to arrive at the desired solution



Training

Supervised Learning

Supervised Learning techniques need external supervision to train models



Unsupervised Learning

Unsupervised Learning techniques do not need any supervision to train models



Reinforcement Learning

Reinforcement Learning techniques do not need any supervision to train models



Type of Problems

Supervised Learning

Classification and Regression



Unsupervised Learning

Clustering and Association



Reinforcement Learning

Reward based



Applications

Supervised Learning



Weather Prediction



Sales Forecast



Stock Price Analysis

Unsupervised Learning



Customer Segmentation



Churn Analysis

Reinforcement Learning



Building Games



Training Robots